Handout for Week 9

Understanding the Concept of Truth Values

Course materials available at: https://spaceofreasons.netlify.app/courses/frege2025/

Plan for unpacking the concept of *truth values*:

1. Three versions of a *basic discursive bipolarity*:
   Semantic: true/false, Pragmatic: accept/reject, and Rational-functional.
2. Two explanatory roles truth values play:
   Dummett's "free-standing" and "ingredient" conceptions of content.
   A notion of extensionality of the relation between them.
3. A formal model: synthetic multivalued logic.
   Distinguishing (ingredient) *multivalues* and (free-standing) *designatedness* values.
4. *Analytic*, substitutional use of the apparatus of multivalued logic.
   Semantic analogues of Lindenbaum algebras.
5. Relations and significance of some actual multivalued logics:
   K3 (Kripke), LP (Graham Priest's "Logic of Paradox"), ST (Strict-Tolerant), and FDE (First Degree Entailment).
6. Relating the *assertional* substitutional hierarchy to the *inferential* substitutional hierarchy by shifting focus of pragmatic account of free-standing (designated) content.
   Another notion of extensionality, from the relations between these substitutional hierarchies.

Frege:
We have seen that it is true of pans of sentences that they have *Bedeutungen.*
What of a whole sentence, does this have a *Bedeutung* too?
If we are concerned with truth, if we are aiming at knowledge, then we demand of each proper name occurring in a sentence that it should have a *Bedeutung.*
On the other hand, we know that as far as the sense of a sentence, the thought, is concerned, it does not matter whether the parts of the sentence have *Bedeutungen* or not.
It follows that there must be something associated with a sentence which is different from the thought, something to which it is essential that the parts of the sentence should have *Bedeutungen.*
This is to be called the *Bedeutung* of the sentence.
But the only thing to which this is essential is what I call the truth-value—whether the thought is true or false….
A sentence proper is a proper name, and its *Bedeutung,* if it has one, is a truth-value: the True or the False. [Beaney 297-8]

Michael Dummett (*Frege's Philosophy of Language*):
In speaking of sentences themselves there are two different ways in which we may regard them; and these may give rise to two distinct notions of [content]. On the one hand, we may think of sentences as complete utterances by means of which, when a specific kind of force is attached, a linguistic act may be effected: in this connection, we require that notion of [content] in terms of which the particular kind of force may be explained. On the other hand, sentences may also occur as constituent parts of other sentences, and, in this connection, may have a semantic role in helping to determine the [content] of the whole sentence: so here we shall be concerned with whatever notion of [content] is required to explain how the [content] of a complex sentence is determined from that of its components. There is no a priori reason why the two notions of [content] should coincide.

Free-standing vs. ingredient content. Two expressions have the same ingredient content iff substituting one for the other never changes the free-standing content of expressions in which they occur. In multivalued logic: designatedness values and multivalues.

Example:

!	**[*1]**	**[2]**	**[3]**
[*1]	1	2	3
[2]	1	3	3
[3]	2	3	1

Three *multivalues*: [1], [2], and [3], of which one, [1], is *designated*, *.
Compute the values of and consequence relations among sentences of the form '**p!q**'.

Kleene's Strong Three-Valued Connective Definitions

A	**¬A**
F	T
U	U
T	F

A∨B	**F**	**U**	**T**
F	F	U	T
U	U	U	T
T	T	T	T

A&B	**F**	**U**	**T**
F	F	F	F
U	F	U	U
T	F	U	T

K3: **T** designated. Truth-value *gaps*.
LP (Logic of Paradox): **T** and **U** designated. Truth-value *gluts*.
Consequence preserves designatedness.
ST (Strict-tolerant):
Consequence holds if strictly true premises don't lead to strictly false conclusions.
FDE (First degree entailment):
Four-valued, with extra values being 'Neither' (true nor false), as in K3
and 'Both' (true and false), as in LP.